

SHIV DUTT CHOUBEY

M.Sc. Cybersecurity Candidate • Cybersecurity Researcher

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SUMMARY

I am a motivated computer science graduate student and researcher specializing in network security, ethical hacking, and cryptography. Currently pursuing my M.Sc. in Cyber Security, I have a strong academic record and research experience at leading institutions. My proficiency in intrusion detection, secure API development, and IIoT security, combined with my commitment to enhancing cybersecurity, positions me as a valuable contributor to this dynamic field.

EDUCATION

M.Sc. Cyber Security — SPI 9.05

National Forensic Sciences University (NFSU), Gandhinagar

Pursuing → Jul 2027

B.Sc. Computer Science — 80 %

Rani Durgawati Vishwavidyalaya (RDVV)

Jul 2024

AISSCE Class XII — CBSE 91 %

Army Public School No. 1, Jabalpur

Jul 2021

AISSE Class X — CBSE 91.2 %

Army Public School No. 1, Jabalpur

Jul 2019

PROFESSIONAL EXPERIENCE

NPTEL Pre-Doctoral Research Fellow

IIT Kharagpur, Kharagpur, West Bengal

Aug 2024 – Jul 2025

- Led research on **Quantum Internet Architecture & Post-Quantum Networking** under sponsored project “Design and Development of Future Quantum Internet Architecture and Post-Quantum Modern Networking.”
- Designed post-quantum cryptographic protocols for next-gen Internet infrastructure.
- Investigated IIoT communication systems; proposed improved frameworks.
- Produced technical reports; collaborated with faculty and peer researchers.

NPTEL Summer Research Intern

RGSOIPL, IIT Kharagpur & K.L.E. Technological University (Remote)

Apr – Jul 2024

- Analysed patent landscapes in AI and software-security domains.
- Conducted patentability searches, prior-art analysis, and IP documentation.

Software Developer Intern

Doions Private Limited, Jabalpur

Nov 2022 – Mar 2023

- Built **Data Processing Pipeline for Conversational AI** for production chatbot (*Local Learn Chatbot*), boosting throughput and response quality.
- Contributed to backend API development, QA, and Agile sprint cycles.

PROJECTS

ERIV — Post-Decryption Semantic Protection vs. HNDL Attacks

- Semantic security layer protecting data *after* decryption against Harvest-Now-Decrypt-Later quantum threats.

Netflow Monitor — Low-Cost IDPS for SMBs

- Lightweight IDPS using network-flow analysis and anomaly detection; minimal false positives.

LoRa ImageSum — Secure Image Transmission over LoRa

- Cryptographic checksums for data integrity over lossy low-power radio.

NLP-AST Compiler Security — Source-Code Integrity

- AST + NLP to detect code manipulation and unauthorised access pre-compilation.

LiveMedQR — Secure Medical EHR Device

- Hardware device transmitting medical unit data to patient device via Wi-Fi; reduces staff errors and improves EHR accuracy.

Keyword Detection API — Content Moderation

- RESTful API flagging keywords for content moderation and threat-intelligence workflows.

CORE COMPETENCIES

- Network Security & IDPS
- Ethical Hacking & Pen Testing
- Cryptography & PQC
- Intrusion Detection Systems
- Secure Coding Practices
- Malware Analysis
- Web & Mobile App Security
- Quantum Computing Research
- IIoT Security
- Natural Language Processing
- API Dev & Integration
- Patent Analysis & IP Research
- Database Management
- Cloud Computing
- Digital Forensics
- Incident Response
- Audit & Compliance

TECHNICAL SKILLS

Languages: Python, Java, C, C++, Kotlin, JavaScript, SQL, Bash

Security Tools: Wireshark, Nmap, Metasploit, ZAP, Frida, Objection

Networking: Packet analysis, Firewall config, VPN, LoRa, Device hardening

AI / ML: TensorFlow, Deep Learning, NLP pipelines, Federated Learning

Cloud & DevOps: Docker, Kubernetes, Git, GPU Acceleration

Dev: RESTful API, Full-Stack Web & Mobile, Agile/Scrum

Databases: MySQL, PostgreSQL, MongoDB

Forensics: Seizure & analysis of electronic evidence

CERTIFICATIONS

Fortinet NSE Training Institute

- NSE 3 — Fortinet Certified Associate** (Network security architecture, Fortinet portfolio)
- NSE 1 & 2 — Fortinet Certified Fundamentals** (Threat landscape, cloud security basics)

NPTEL / Indian Institutes of Technology

- Ethical Hacking — IIT-KGP
- Applied AI — IIT Palakkad
- Computer Networks & IP — IIT-KGP
- Industry 4.0 & IIoT — IIT-KGP
- Introduction to IoT — IIT-KGP
- Database Systems — IIT Madras
- DBMS — IIT-KGP
- Programming in Java — IIT-KGP
- Programming in Modern C++ — IIT-KGP
- Introduction to C++ — IIT Bombay
- Introduction to Operating System — IIT Madras
- Computer Graphics — IIT Madras
- Corporate Finance — IIT-KGP
- IP Portfolio Management — IIT Roorkee
- Roadmap for Patent Creation — IIT-KGP

COMPETITIONS & HACKATHONS

- 5G Innovation Hackathon 2026** — **Selected for National Level** by Dept. of Telecom, Govt. of India.
- Anti-Forensics Hackathon** — Identified VeraCrypt hidden volumes; advanced digital forensics.
- File Forensics Bootcamp** — Reconstructed fragmented distributed files from forensic disk images.
- Cyber Security Presentation** — Phishing & social engineering research. **1st Position.**
- Innovate India Coding Championship** — DSA in C++. **National Rank 7,845.**

VOLUNTEER WORK & LEADERSHIP

- WeSeSo (We Secure Society)** — Cybersecurity Volunteer, Apr 2020–Present: digital safety outreach & community programmes.
- NSS, Govt. of India** — Student Volunteer, Nov 2021–Jul 2024: rural development, health awareness drives.
- NCC, India** — Army Wing Cadet, Jul 2016–Jul 2018: military training, leadership & discipline.